Quantum-Ready Tug Scheduling for Tidal Ports

Every harbor tug costs about $2M a year, and every tug held back "just in case" lost revenue handed to a competitor. We framed this daily trade-off as a quantum optimization problem and laid out an path from today's classical tools to quantum.

The problem

Large ships need tugs (small powerful boats) to push and pull them safely to the dock. A tug operator owns only a handful of these boats and must decide daily which tug serves which ship, and when. In tidal ports like Hamburg, Germany, the biggest ships can only enter in short windows around high tide. Wind raises the number of tugs every ship needs at the same moment, and crews have mandatory rest periods. Contracted ships must confirm at least 2 hours ahead, so tugs are held in reserve for them. Servicing uncontracted ships are extra revenue, but only if a tug is free. Challenge: Hold back too many tugs and spot jobs go to competitors, hold back too few and contracts are broken. Today a manager plans all of this by hand on a Gantt chart every morning, and nobody has measured how good those plans are.

Who it affects and why it matters

Harbor towage operators running 5 to 50 tugs in ports where 2 to 4 competitors share the traffic, and the operations planners who build the daily schedule. Figures from the client:

Lever	Value per year	Basis
Reduce fleet by one tug	~$1.95M	$1.5M opex plus $9M purchase price spread over 20 years
One extra spot job per day	~$1.8M	365 jobs × ~$5k average fee

For a 6-tug operator with about $9M in yearly operating cost, either lever is worth roughly 20% of the cost base.

Why quantum

Every decision is a yes/no choice (does this tug serve this ship, do we take this spot job), and the choices interact. The number of combinations grows exponentially with the port, which is the structure quantum optimization targets. We formulated the problem as a QUBO for QAOA in Qiskit. A QUBO is also the native input for quantum annealers, so the same model can be tested on both types of quantum hardware.

Today’s limits are noisy hardware, and a real port needs thousands binary variables while current machines handle small problems well. Classical solvers handle this problem today, so there is no guaranteed speedup.

We expect to see a payoff for large hub ports and planning for many weather and confirmation scenarios at once instead of a single forecast (tug operators in peak-driven ports like Port Miami, Port Everglades, or Port Canaveral could also benefit). That multiplies the problem size and is

where classical solvers struggle most. It requires fault-tolerant hardware, which public roadmaps place in the late 2020s at the earliest.

Next steps

1. Collect: historical schedules, weather, tides and crew rosters from the operator
2. Benchmark: solve each past day with a classical solver (OR Tools) and express the gap to the manual plan in $
3. Verify quantum: run the same QUBO on gate-based hardware (QAOA) and on a quantum annealer (e.g., D-Wave), confirm that both find the OR-Tools optimum on small cases, and compare which scales better as cases grow
4. Forecast: train ML models to predict arrivals, confirmations and tug demand
5. Plan daily: forecasts feed the same optimizer used in the benchmark; the tug operator's planner approves
6. Switch: move to QC once it consistently performs better

After the switch, only the solver changes: ML forecasts become the day's problem, a quantum optimizer solves it with OR-Tools as fallback, a rule check confirms the plan, the planner approves, and the plan is re-solved in minutes when wind or confirmations change.